```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
// Base Class
class User {
protected:
   string name;
   int age;
   string skinType;
public:
   User(string n, int a, string s) {
      name = n;
      age = a;
      skinType = s;   }
   virtual void printInfo() {
      cout << "\n=======================================\n";
      cout << "            USER INFORMATION          \n";
      cout << "=======================================\n";
      cout << left << setw(15) << "Name:" << name << endl;
      cout << left << setw(15) << "Age:" << age << endl;
      cout << left << setw(15) << "Skin Type:" << skinType << endl;
   }
   // Pure virtual function for polymorphism
   virtual void recommendMakeup() = 0;
   virtual ~User() {} };
// Derived Class 1
class WarmUndertone : public User {
public:
   WarmUndertone(string n, int a, string s) : User(n, a, s) {}
```

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    void printInfo() override {
        User::printInfo();
        cout << left << setw(15) << "Undertone:" << "Warm" << endl;
    }
    void recommendMakeup() override {
        cout << "\n=======================================\n";
        cout << "      MAKEUP RECOMMENDATIONS           \n";
        cout << "=======================================\n";
        cout << left << setw(15) << "Foundation:" << "Yellow or golden undertone foundation" << endl;
        cout << left << setw(15) << "Blush:" << "Peach or coral blush" << endl;
        cout << left << setw(15) << "Lipstick:" << "Warm nude, terracotta, or brick red" << endl;
        cout << left << setw(15) << "Eyeshadow:" << "Gold, bronze, or warm brown tones" << endl;
    }
};
// Derived Class 2
class CoolUndertone : public User {
public:
    CoolUndertone(string n, int a, string s) : User(n, a, s) {}
    void printInfo() override {
        User::printInfo();
        cout << left << setw(15) << "Undertone:" << "Cool" << endl;
    }
    void recommendMakeup() override {
        cout << "\n=======================================\n";
        cout << "      MAKEUP RECOMMENDATIONS           \n";
        cout << "=======================================\n";
        cout << left << setw(15) << "Foundation:" << "Pink or rosy undertone foundation" << endl;
        cout << left << setw(15) << "Blush:" << "Pink or berry blush" << endl;
        cout << left << setw(15) << "Lipstick:" << "Rose, berry, or blue-based red" << endl;
        cout << left << setw(15) << "Eyeshadow:" << "Silver, mauve, or cool brown tones" << endl;
    }
```

```
};
// Derived Class 3
class NeutralUndertone : public User {
public:
   NeutralUndertone(string n, int a, string s) : User(n, a, s) {}
   void printInfo() override {
      User::printInfo();
      cout << left << setw(15) << "Undertone:" << "Neutral" << endl;
   }
   void recommendMakeup() override {
      cout << "\n=======================================\n";
      cout << "      MAKEUP RECOMMENDATIONS           \n";
      cout << "=======================================\n";
      cout << left << setw(15) << "Foundation:" << "Neutral beige foundation" << endl;
      cout << left << setw(15) << "Blush:" << "Soft pink, peach, or rose blush" << endl;
      cout << left << setw(15) << "Lipstick:" << "Nude, rose, or soft red" << endl;
      cout << left << setw(15) << "Eyeshadow:" << "Champagne, taupe, or neutral brown tones" <<
endl;
   } };
int main() {
   string name, skinType;
   int age;
   int choice;
   cout << "Enter your name: ";
   getline(cin, name);
   cout << "Enter your age: ";
   cin >> age;
   cout << "Enter your skin type ";
   cout << "(dry / oily / combination): ";
   cin >> skinType;
```

```
cout << "\nChoose your skin undertone:\n";
cout << "1. Warm\n";
cout << "2. Cool\n";
cout << "3. Neutral\n";
cout << "Your choice: ";
cin >> choice;
User* user = nullptr;
if (choice == 1) {
   user = new WarmUndertone(name, age, skinType);  }
else if (choice == 2) {
   user = new CoolUndertone(name, age, skinType);  }
else if (choice == 3) {
   user = new NeutralUndertone(name, age, skinType);   }
else {
   cout << "\nInvalid choice. Program ended.\n";
   return 0;   }
user->printInfo();
user->recommendMakeup();
delete user;
return 0;}
```